

Matchete vs Fixed Einstein-Hilbert Comparison

ORBIT

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1 Setup and Conventions

This report compares a Matchete-derived Lagrangian to a fixed reference action

$$\mathcal{L}_{\text{ref}} = \Lambda_{\text{ref}} \sqrt{-g} + \frac{2}{\kappa_{\text{ref}}^2} \sqrt{-g} R \quad (1)$$

with fixed values $\Lambda_{\text{ref}} = 1$ and $\kappa_{\text{ref}} = 1$. The Matchete input is normalized by setting $\hbar = 1$ and discarding all terms containing negative powers of ϵ .

2 Reference Support Through Mass Dimension 6

The supported EH plus cosmological sectors are $\{n_h, 0\}$ for $1 \leq n_h \leq 6$ together with $\{2, 2\}$, $\{3, 2\}$, and $\{4, 2\}$.

3 Reduced Matchete Input

Sector	Input terms	IBP terms	Reduced terms	Coordinates
1, 0	3	3	3	1
2, 0	6	6	6	2
2, 2	24	12	12	4
2, 4	26	10	10	5
3, 0	9	9	9	3
3, 2	77	39	18	10
4, 0	15	15	15	5
4, 2	182	90	75	31
5, 0	21	21	21	7
6, 0	33	33	33	11

4 Reduced Fixed Reference

Sector	Input terms	IBP terms	Reduced terms	Coordinates
1, 0	1	1	1	1
1, 2	2	0	0	0
2, 0	2	2	2	2
2, 2	10	4	4	4
3, 0	3	3	3	3
3, 2	28	13	6	10
4, 0	5	5	5	5
4, 2	66	38	25	31
5, 0	6	6	6	7
6, 0	9	9	9	11

5 Sector-by-Sector Comparison

Sector	Input coords	Reference coords	Difference zero
1, 0	1	1	no
2, 0	2	2	no
2, 2	4	4	no
3, 0	3	3	no
3, 2	10	10	no
4, 0	5	5	no
4, 2	31	31	no
5, 0	7	7	no
6, 0	11	11	no

5.1 Unsupported sectors

$\{2, 4\}$

5.2 Supported-sector matching equations

$\{2*M^4*\backslash[Kappa] - 8*Scalar[\backslash[Epsilon]] + 3*M^4*\backslash[Kappa]*Scalar[\backslash[Epsilon]] + 2*M^4*\backslash[Kappa]*Log[Scalar[\backslash[Mu]bar2]/Scalar[M]^2] + (2*M^4*\backslash[Kappa]*Log[Scalar[\backslash[Epsilon]]])\}$

5.3 Solved constraints

$Reduce[\{2*M^4*\backslash[Kappa] - 8*Scalar[\backslash[Epsilon]] + 3*M^4*\backslash[Kappa]*Scalar[\backslash[Epsilon]] + 2*M^4*\backslash[Kappa]*Log[Scalar[\backslash[Mu]bar2]/Scalar[M]^2] + (2*M^4*\backslash[Kappa]*Log[Scalar[\backslash[Epsilon]]])\}]$

6 Differences

6.1 Supported difference

$(h[a, -a]*(-8 + 3*M^4*\backslash[Kappa] + 2*M^4*\backslash[Kappa]*Log[Scalar[\backslash[Mu]bar2]/Scalar[M]^2] + (2*M^4*\backslash[Kappa]*Log[Scalar[\backslash[Epsilon]]])\})$

6.2 Unsupported difference

$-1/80*(\backslash[Kappa]^2*(1 + Log[Scalar[\backslash[Mu]bar2]/Scalar[M]^2]*Scalar[\backslash[Epsilon]])*PD[c][PD[b][h[a, -a]]])$

7 Conclusion

No exact full match was established under the current fixed-reference conventions. The unsupported-sector content and/or the solved supported-sector equations obstruct equality.

8 Validation Summary

$\langle | "TranslationMatchesBaselineQ" \rightarrow True, "TranslationDifference" \rightarrow 0, "NormalizedContainsNoHb" \rangle$